LUND UNIVERSITY

Written Examination
Linear Analysis, MATB24
April 17, 2021
Time: 09.00–16.00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Hand in electronically in the form of a PDF-file no later than at 15.59. You may freely use formulas from the formula sheet on the canvas page. Refer to relevant theorems from the course when you use them. Write clearly with short and concise motivations.

1. Find a solution $u(x,t)$ to the heat equation $u_{xx} = u_t$ when $t > 0$, $0 < x < \pi$ such that the boundary condition $u_x(0,t) = u_x(\pi,t) = 0$ and the initial condition $u(x,0) = \sin^2 x$ are satisfied.

2. a) Determine the pointwise limit $f(x) = \lim_{n\to\infty} f_n(x)$ for $x \geq 0$ where

$$f_n(x) = \frac{nx + n^2x^3}{1 + n^2x^2}.$$

b) Investigate whether or not the convergence is uniform on $[0,\infty)$.
c) Recall that f_n is said to converge to f in L^2-sense if $\|f - f_n\|_2 \to 0$ as $n \to \infty$ where $\|f\|_2^2 = \int_0^\infty |f(x)|^2\,dx$. Determine whether or not L^2-convergence holds in the present case.

3. Determine the Fourier series of the 2π-periodic function $f(x)$ which equals to $x^3 - \pi^2 x$ when $|x| \leq \pi$. Also determine the sum $\sum_1^\infty \frac{1}{n^6}$.

4. Let $f(x) = xe^{-x}$ for $x \geq 0$ and $f(x) = 0$ for $x \leq 0$.
a) Find the Fourier transform of f.
b) For each $\lambda > 0$ find the value of the integral

$$\int_{-\infty}^{+\infty} \frac{\sin \lambda x}{x(1+ix)^2}\,dx.$$

5. a) Let $g(x) = e^{-|x|}$. Find the convolution $g * g(x) = \int_{-\infty}^{\infty} g(x-t)g(t)\,dt$.
b) Use Fourier transforms to solve the differential equation $y''(x) - y(x) = g(x)$, $x \in \mathbb{R}$.

6. For a positive integer n, we define $K_n(x,y) = \frac{1}{\pi} n e^{-n(x^2+y^2)}$.
(a) Prove that K_n is a *positive summation kernel on* $\mathbb{R}^2$, i.e.,:
(i) $K_n(x,y) \geq 0$ for all $(x,y) \in \mathbb{R}^2$ and all n,
(ii) $\iint_{\mathbb{R}^2} K_n(x,y)\,dxdy = 1$ for all n,
(iii) for each $\delta > 0$, $\lim_{n\to\infty} \iint_{\{(x,y)\in\mathbb{R}^2\,;\,x^2+y^2>\delta^2\}} K_n(x,y)\,dxdy = 0$.
(b) Now fix a continuous and bounded function $f : \mathbb{R}^2 \to \mathbb{C}$. Try to figure out the limit

$$\lim_{n\to\infty} \iint_{\mathbb{R}^2} f(x,y)K_n(x,y)\,dxdy.$$

Motivate your statement as carefully as possible.